

A Review of Cyclic Olefin Copolymer Applications in Microfluidics and Microdevices

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Cyclic olefin copolymers (COC) are amorphous, transparent thermoplastics composed of cyclic olefin monomers (norbornene) and linear olefins (ethene). They are increasingly utilized as fabrication materials for microsystems and microfluidic devices, owing to their promising features of low water absorption, high electrical insulation, long-term stability of surface treatments, and resistance to a broad variety of acids and solvents. Many manufacturing processes for COC-based devices have been developed in recent decades. These methodologies are categorized as replication methods or fast prototyping as common in fabrication of thermoplastic microfluidic devices. This review gives a full discussion of the features of COCs, the various production processes, and the numerous selected applications in microfluidic platforms. The review also explores COC's composition and fundamental features, as well as fabrication processes and applications in a variety of fields, investigates the material's potential advantages and uses, and attempts to create a comprehensive list of COC's possible benefits. Due to their unique features and simplicity of fabrication, COCs are projected to advance the future of microfluidics, microsystems, and optofluidics.

utilizes hollow structures called microchannels implanted in a chip to provide efficient fluid flows.^[1] A microfluidic platform's average overall size is in the range of a few square millimeters to centimeters, with a characteristic dimension of less than 1 mm and a fluid handling capacity of microliters (10^{-6}) to picoliters (10^{-12}) (and sometimes femtoliters (10^{-15})). LOC is a term that refers to microdevices that combine a variety of laboratory procedures onto a single chip. Micro-total-analysis-systems (TAS) are a closely related family of devices that are usually regarded as combining all laboratory operations necessary for analysis on a single platform. Due to a minor distinction between them, the words "LOC" and "TAS" are often used interchangeably.^[2] Nonetheless, these platforms are predicted to change the analytic and diagnostic disciplines just like integrated circuits revolutionized the electronics and communication industries.

1. Introduction

A key driving force in microfluidics technology is the advancement of lab-on-a-chip (LOC) devices. Microfluidics technology

Material selection is important to the TAS/LOC platforms' fabrication approach as well as operational performance. Indeed, with the introduction of novel biocompatible materials and enormous breakthroughs in microfabrication technologies, the TAS and LOC platforms

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DOI: 10.1002/mame.202200053

have already shown their superiority in a variety of applications when compared to their traditional macro-scale equivalents. Their superior characteristics include, but are not limited to, increased sensitivity and selectivity, reduced sample usage and wastage, reduced processing time, accuracy, cost-effectiveness, configurability, and the ability to operate autonomously and repeatedly, thereby eliminating human error.^[3,4]

For many years, silicon and glass were the predominant substrate materials for manufacturing these platforms.^[5] The reason for their prevalence is that these materials inherited quite well-established processing techniques from the microelectronics industry.^[6] As is well known, silicon and glass wafer fabrication techniques such as photolithography, dry/wet etching chemistries, and packaging have matured significantly over the decades and are highly optimized for large-scale production. Silicon is a preferred substrate material because of its availability, semiconducting properties, and relative ease of surface modification. It has excellent integration capabilities with metallic electrodes and other materials such as polydimethylsiloxane (PDMS). Its ability to withstand elevated temperatures and excellent thermal conductivity makes it an appropriate choice for applications requiring fast thermal cycling, such as polymerase chain reaction (PCR).^[7] However, it has its own limitations: Silicon is a relatively expensive material and requires a cleanroom and specialized equipment for the fabrication of devices.^[8] Moreover, it is not transparent and hence cannot be employed in applications that involve optical detection.^[9] Furthermore, owing to its high modulus of elasticity, silicon is quite fragile. Therefore, its integration with other active components such as pumps and valves is not a straightforward task.^[7]

Glass also offers numerous advantages. First, it is biocompatible, which allows it to be used in a broad range of biological applications; this is the reason it has been used in a number of LOC platforms.^[10,11] Second, the surface of glass can also be easily functionalized; therefore, it is extremely useful in applications such as biosensing or those requiring material coating to minimize nonspecific adsorption. Third, the excellent thermal stability and chemical inertness of glass make it an appropriate choice for processes involving high temperatures and a variety of solvents. This also enables effective washing with the help of either chemicals or heat. Hence, glass-based devices can be used multiple times. In contrast to silicon, glass is optically transparent in the visible as well as UV range making it a natural choice for optical detection. Additionally, glass is relatively cheaper than silicon; however, its mass production into chips is still expensive. Furthermore, glass is an electrical insulator that can integrate metal electrodes and active components with relative ease. Nevertheless, glass also sometimes requires cleanrooms during device microfabrication which consumes additional time, specialized equipment, labor, and time.^[8,12]

Due to the limitations of silicon and glass, the search for alternative substrate materials for μ TAS/LOC systems shifted the focus to polymer-based substrates. Polymers are significantly less expensive than both silicon and glass. They do not rely on serial manufacturing methods used in the cleanroom, implying easier fabrication methods and mass production suitability at lower costs.^[12] PDMS is one of the most often used polymers in microfluidic devices,^[13] an elastomer compatible with the standard microfluidic fabrication process. It is inexpensive, and easy

to mold, exhibiting optical transparency, gas permeability, hydrophobicity, biocompatibility, low autofluorescence, and high elasticity. As a result, PDMS is an advantageous material for biological research applications such as long-term cell culture and biochemical testing. In addition, thin PDMS membranes can also be used for valves and pumps in microchannels. However, PDMS has its limitations. For instance, it is incompatible with several organic solvents (e.g., acetone, methanol, and chloroform), due to the fact that their molecules can be adsorbed onto the channel walls, therefore swelling the microchannel. Evaporation of water vapor through channel walls is also a problem as it may lead to a change in the concentration of an aqueous solution in the channel, thus changing experimental conditions and results. Finally, the PDMS surface is inherently hydrophobic and requires surface treatment, most commonly oxygen plasma, to convert its hydrophobicity to hydrophilicity. However, there is a tendency of reversion to the native hydrophobic nature. Thus, the device needs to be used within hours after the surface treatment which limits its mass production for later usage.

Other polymeric materials for the fabrication of microfluidic devices include poly(methyl methacrylate) (PMMA),^[14] polycarbonate (PC),^[15] cyclic olefin polymers (COP),^[16] cyclic olefin copolymers (COCs),^[16,17] and Thiol-ene polymers (TEs).^[18] COCs are a thermoplastic copolymer composed of ethylene and norbornene (NB) that is catalyzed with metallocene.^[19] They are of special interest because of their unique properties compared to other materials. Currently, COC is a registered trademark of and manufactured by Topas Advanced Polymers.^[19] Topas is an abbreviation for thermoplastic olefin polymer of amorphous structure. Apart from Topas, COCs are commercially available from APEL by Mitsui Chemicals America Inc.^[20]

COC has a number of advantages. For example, COC's advantageous properties include a low water absorption rate, good electrical insulating properties, and biological inertness. As a result, COC is widely used in biological,^[21,22] membrane,^[23,24] and semiconductor^[25,26] applications. The surface treatments have been shown to be stable over time, and the material is resistant to polar solvents and acids. The material is attacked by only non-polar organic solvents such as cyclohexane or toluene. Additionally, it exhibits great stiffness and optical transparency, especially in the UV spectrum. The heat deflection temperature (the temperature at which a polymer deforms under the effect of a certain load) varies between 70 and 170 °C depending on the COC's grade. Numerous fabrication techniques, such as micromilling, injection molding, and hot embossing, are available for creating microchannels in COC.^[19] COCs in microfluidics began in the early 21st century as a candidate for μ TAS/LOC.^[27,28] Initial publications focused on fabrication demonstration and feasibility utilizing mainly hot embossing, and injection molding.^[29–31] With the advancements in fabrication processes such as stamp fabrication, and bonding techniques, COC-LOC applications were possible including *Escherichia coli* detection,^[32] sperm count determination,^[33] and a COC point-of-care device integrated with a smartphone to detect silicosis and perform mobile analysis.^[34]

COPs are closely related to COCs and share many of the advantages mentioned previously, however, they differ in the synthesis process. COPs are created by the ring opening metathesis polymerization of cyclic monomers, followed by hydrogenation.^[35] This polymer is available commercially by: Zeonex (Nippon

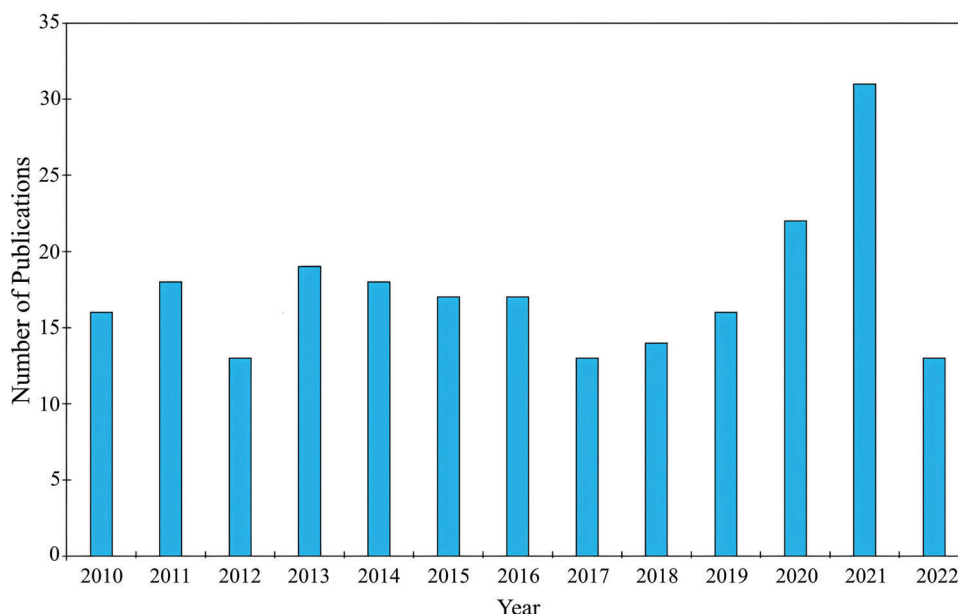


Figure 1. Number of publications on “Cyclic Olefin Copolymer” during the period 2010–2022 (for year 2022, the papers published until March 16, 2022 are included).

Zeon, Japan)^[36] and JSR Corporation (Japan).^[37] Other than the synthesis process, COPs have a relatively higher moisture absorption, smaller range of glass transition temperatures (T_g),^[38] higher cost due to its more complex synthesis, and COCs have higher flowability which results in better injection molding.^[39]

To the authors' best knowledge, the only comprehensive review of COC based microfluidics were published in 2006^[29] and 2010.^[16] However, in the former article, COC was a relatively new material to microfluidics and in the latter one, there was no distinction made between COC and COP microchannels. The authors believe that such conflation between the two polymers may be misleading to readers seeking material selection strategies. Thus, in this review, COC is the main focus and COP will be stated clearly when used. The majority of existing reviews focused mainly on rigid thermoplastics such as PMMA, PS, and PC,^[40–46] and flexible polymers^[47] such as parylene, and poly(ethylene terephthalate) for biomedical application. Others reviewed the fabrication aspects of polymer microfluidics,^[48,49] and the bonding of various thermoplastics and PDMS.^[50–52] Furthermore, a search of databases (Crossref, Google Scholar, ScienceDirect, and Web of Science) for the keyword “cyclic olefin copolymer” reveal that there has been a significant increase in the number of publications involving COC over the past decade as shown in **Figure 1**. Hence, there is still much room to investigate the material's potential advantages and applications. The current review article presents the composition and fundamental properties, fabrication and bonding techniques, and applications of COC in numerous fields. The remaining sections of this article are organized as follows: Section 2 discusses the physical and chemical properties of COC. Section 3 provides a detailed account of the fabrication methods used for COC-based microdevices. In Section 4, the different techniques available for COC surface modification are described. Section 5 details the three main COC bonding methods. Section 6 discusses the application areas

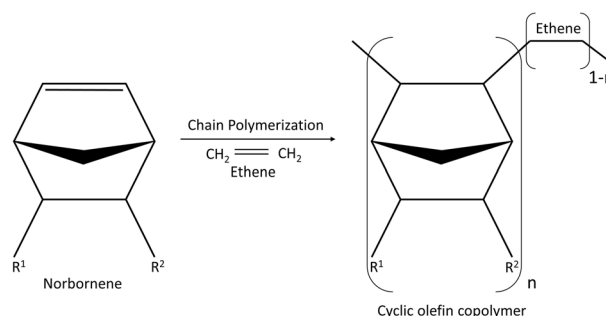


Figure 2. Chain copolymerization of norbornene with ethene.

where COC has been used extensively in recent years. Finally, Section 7 provides the conclusion and indicates fruitful future research directions involving the COC material.

2. Chemical and Physical Properties of COC

COCs are amorphous, transparent copolymers of cyclic olefin monomers (NB) and linear olefins (ethene).^[53] Two main COC brands are available commercially: TOPAS by Topas Advanced Polymers GmbH^[54] and APEL by Mitsui Chemicals America Inc.^[55] Generally, COCs are prepared by the chain copolymerization of a cyclic olefin with an olefin as shown in **Figure 2**. Where TOPAS manufacture COCs using NB (cyclic olefin) and ethene (olefin) while APEL use tetracyclododecane as the cyclic olefin.^[56]

2.1. Chemical Properties

2.1.1. Surface Energy

COC wettability is of great interest when employing COC as a substrate for microfluidic devices. Wettability is quantified in

Table 1. Chemical resistance comparison between COC and other microchannel material.^[54,55,61]

Solvent	COC	PMMA	PC	PS	PDMS
Acid-dilute	r	d	r	r	r
Acid-concentrated	r	d	d	d	r
Alkalies	r	s	d	r	r
Alcohols	r	d	r	r	d
Aliphatic hydrocarbons	d	r	r	d	r
Aromatic hydrocarbons	d	d	d	d	s
Esters	r	d	d	d	n
Halogenated hydrocarbons	d	d	d	d	d
Ketones	s	d	d	d	d
Aldehydes	s	s	s	d	n

(r) indicates resistant, (s) swelling, (d) non-resistant (dissolves), and (n) no available data.

terms of contact angle. When the water contact angle is between 0° and 90°, the surface is designated as being hydrophilic and is characterized by having a high surface energy, and when the contact angle is greater than 90°, the surface is designated as being hydrophobic with a low surface energy.^[57] Native COC is hydrophobic with a contact angle in the range of 90°–100° (depending on the supplier and grade), making it useful in applications that require water vapor barriers. In addition, water absorption is a key property for materials used in microfluidics, especially in biomedical applications where many of the solvents are aqueous. COCs have low water absorption (<0.01%), about five times less than PDMS and 10–30 times less than PMMA.^[54,55] Low water absorption ensures microchannel dimensional stability in aqueous environments. This is important in passive microchannels where geometry is the main limiting factor in the device.^[58] On the other hand, COC's hydrophobicity may impede its incorporation in biological applications where many of the solvents are aqueous.^[59] However, there are several chemical and physical methods developed to modify the surface wettability of COCs^[60] that will be discussed in Section 4.

2.1.2. Chemical Resistivity

COCs also exhibit high resistivity to acids, bases, and alcohols. However, it dissolves in non-polar organic solvents such as hexadecane and toluene, and swells in aldehydes such as formaldehyde.^[61]

Table 1 compares the chemical resistance between COC and other commonly used materials in microchannels. Its chemical properties make it a suitable substrate in photolithography where it can withstand processes such as etching, patterning, and developing.

2.1.3. Thermal Properties

COC thermal properties depend on the commercial producer and grade. Higher NB content results in COCs with higher glass transition temperatures (T_g). T_g is a characteristic temperature below the melting point where thermoplastics exhibit softening; however, they return to their original state upon cooling.^[60] **Figure 3**

shows different COC grades offered by TOPAS and the proportional relationship between T_g and the NB content, ranging from 80 to 175 °C. This allows COCs to be incorporated in a wide variety of applications without the risk of significant deformation. The softening of the COC at the T_g is an important feature that facilitates hot embossing fabrication, and thermal bonding, as will be discussed in Section 3. **Figure 4** compares the T_g of COC (TOPAS and APEL) with other polymers, indicating some grades of COC exhibit higher T_g than PMMA, PC, and PS. Singh and Mellinger^[62] reported the thermal diffusivity calculations of commercially available thin polymer films in the thickness direction using the thermal wave method technique. They reported thermal diffusivity for two commercially COC grades (TOPAS 6013 and TOPAS 8007) for samples having a thickness of 29 and 13.5 μm. Thermal diffusivity values of $(14.1 \pm 0.02) \times 10^{-7}$ and $(11.7 \pm 0.02) \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$ were reported for both grades, respectively. They were in good agreement with the measurements provided by the manufacturer.

2.2. Physical Properties

2.2.1. Optical Properties

Recently, COCs have become an attractive material for optical and transmission applications. Hessler et al.^[63] incorporated a COC substrate into an optofluidics device for biosensing. Chen et al.^[64] exploited COC substrates as waveguides operating from 26 to 40 GHz, and Zhong et al.^[65] utilized a thin-walled COC tube for a terahertz (THz) waveguide application. These applications are possible because of COCs' high light transmission over a wide range of wavelengths, high refractive index, low birefringence, and large Abbe number.^[54] **Figure 5** shows the light transmission of APL5014 COC (APEL) from UV to IR wavelengths, with high transmission between 350 and 1100 nm. In addition, **Figure 6** shows the UV transmission of various thermoplastics. COC has a high degree of transmission over a wider range in comparison to PMMA, PC, and PS. Denz et al.^[66] combined microfluidics and X-ray technology for biomedical applications. X-ray microfluidic devices have specific requirements that include X-ray resistance, transparency, and low background signal scattering. Materials such as PDMS, silicon,^[67] stainless steel, and kapton film^[68] are commonly used for such applications. However, COC exhibits better X-ray properties in comparison to PDMS and silicon, such as low photon absorption. Furthermore, another important optical property to consider is autofluorescence. Materials under consideration must exhibit low autofluorescence when used in applications such as fluorescence activated cell sorting for optimum optical measurements. COC exhibits higher autofluorescence than PDMS but similar values to PMMA and PC.^[69] In **Table 2**, the optical properties of COCs are compared to those of PMMA, PC, PDMS, and PS.

2.2.2. Mechanical Properties

Similar to the thermal properties, COCs' mechanical properties vary slightly depending on the different grades and the ratio of the cyclic olefin to the olefin used during chain copolymerization, where COC grades with higher NB content exhibit higher